Workflow and architecture patterns for working together 1: Workflows

Jacob Archambault

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Outline

1. Current challenges and their canonical solutions

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2. DevOps 201
   - Communication
     - Conway's Law
     - The Mythical Man Month
   - Throughput
     - Just-in-time manufacturing
     - Goldratt's theory of constraints
     - DevOps Research and Assessment (DORA)

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3. Branching and continuous integration
4. Communication and throughput anti-patterns and solutions
   - Communication anti-pattern: local optimization
   - Throughput anti-pattern: Gandalf vs. the Balrog

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1967: Conway's law

'Organizations which design systems [...] are constrained to produce designs which are copies of the communication structures of these organizations.' - Melvin Conway, 'How do Committees Invent?' Datamation, 1967

Conway's law: examples

- 'If you have four groups working on a compiler, you'll get a 4-pass compiler' - The New Hacker's Dictionary, 1996

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- front-end [devs] business layer [backend devs] monolithic database [DBA team]
- a web api controller [manager] delegates most business logic to business classes [developers] which are part of the same in-memory process [team], while serving as a single entry-point for wider cross-network [cross-team] communication.

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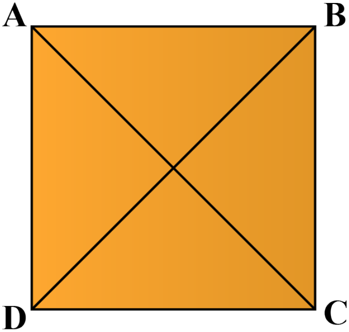
  - branching-based strategies for team management compromise between the above approaches without their benefits

1975: The Mythical Man Month, Fred Brooks

- number of direct communication paths for n individuals$=$ $n(n-1)/2$



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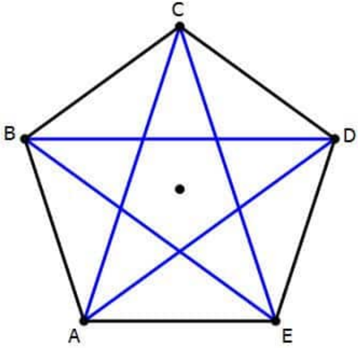

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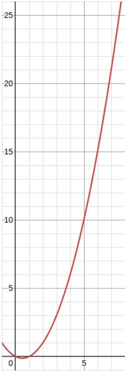
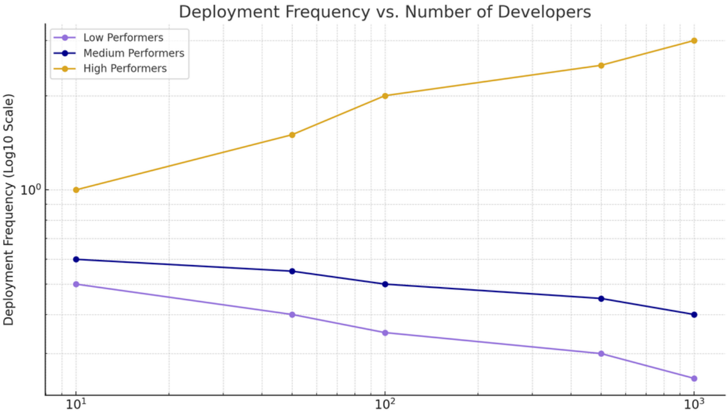


The Mythical Man Month, Fred Brooks (cont.)

- corollary: adding more people to a project can lead not only to diminishing returns on delivery speed, but to objectively less work being completed



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Just-in-time manufacturing and Goldratt's theory of constraints

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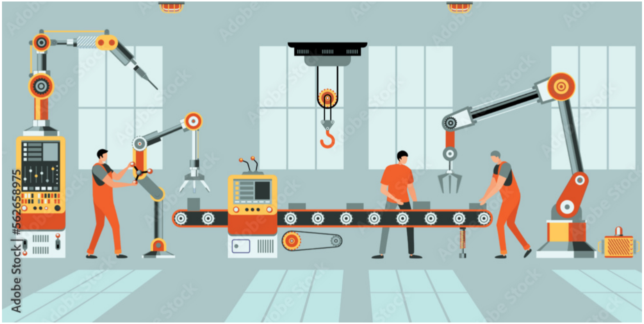
- Remove bottlenecks

Goldratt's theory of constraints (cont.)



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- Uses *cluster analysis* to discover groupings - has no prior understanding of what counts as good or bad.

The State of DevOps Report: throughput metrics

- lead time for changes

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- deployment frequency

The State of DevOps Report: stability metrics

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The State of DevOps Report: results

'Astonishingly, these results demonstrate that there is no trade-off between improving performance and achieving higher levels of stability and quality. Rather, high performers do better at all of these measures. This is precisely what the Agile and Lean movements predict, but much dogma in our industry still rests on the false assumption that moving faster means trading off against other performance goals, rather than enabling and reinforcing them.' - Nicole Forsgren, Jez Humble, and Gene Kim, Accelerate (2017), p. 14

The State of DevOps Report: results

When compared to low performers, elite performers realize

127x faster lead time	**182x** more deployments per year	**8x** lower change failure rate	**2293x** faster failed deployment recovery times

- 2024 State of DevOps Report

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Continuous integration: trade-offs

Continuous Integration applies a different trigger for integration - you integrate whenever you've made a hunk of progress on the feature and your branch is still healthy. There's no expectation that the feature be complete, just that there's been a worthwhile amount of changes to the codebase. The rule of thumb is that "everyone commits to the mainline every day", or more precisely: you should never have more than a day's work sitting unintegrated in your local repository.

- Martin Fowler

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- punishes untracked work (e.g. thorough code reviews/dev testing, helping blocked colleagues)

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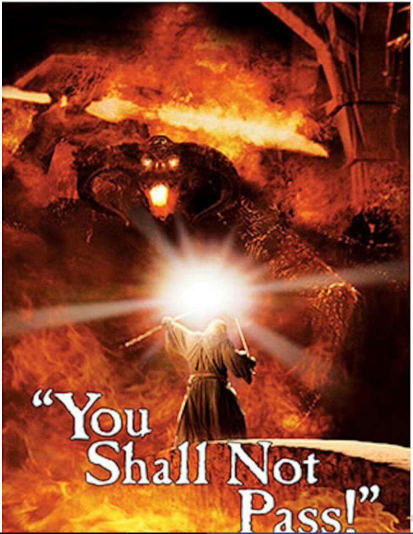

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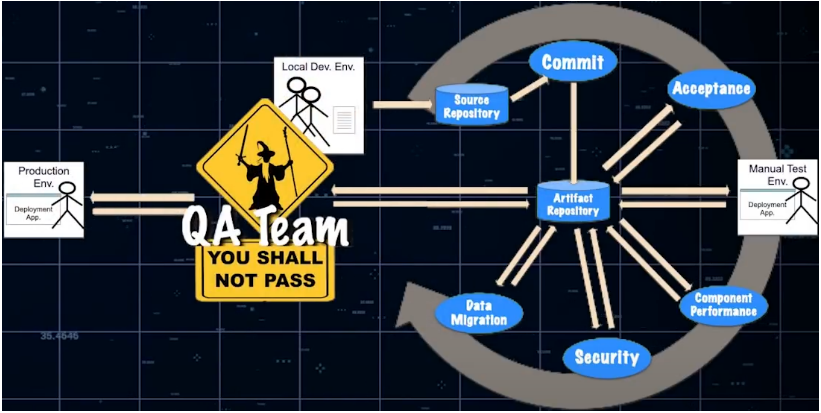

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- these all increase lead time

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- That's mostly it

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'DevOps is whatever you do to bridge friction created by silos, and all the rest is engineering' - Patrick Debois, Puppet State of DevOps report 2021

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